

ENMT301 IMU Assignment 2026

The assignment is to be done on your own and is worth 5% of your ENMT301 grade. There are two parts. Part A is to study the signals output by a [Bosch BNO055 IMU](#) while it was stationary (compared to the lab frame of reference).

Part B is to estimate the path taken by a simulated mobile robot using the IMU data and odometry. The robot moves in straight lines (along its x-axis) and only turns on the spot (around its z-axis). The robot is instrumented with a simulated IMU (similar to the BNO055) that records accelerations and angular speeds.

Each IMU sensor has noise modelled with two components: a Gaussian random-walk process that produces a time varying bias and an additive white Gaussian noise process.

By integrating the angular speeds you can estimate the robot's heading angle. However, you will find that the accelerometer bias produces poor estimates of how far the robot has moved. Instead, a simulated odometer measures how far the robot has moved. Your challenge is to estimate the path followed by the robot using the odometry data with the integrated gyroscope data.

The odometry data has been corrupted by interference. So first you need to determine the interference and apply a digital filter to remove it. Then you can estimate the path the robot took.

The assignment uses a Jupyter notebook running Python. These are popular with engineers and scientists as a useful means of combining text, program code, and program output (plots, etc.) in a single file. You can run the notebook on your own computer or on a Mechatronics / CAE lab computer (including remote access), or online.

You need to submit a short report (using the specified template), showing your plots and describing your observations.

For any problems, post to the [IMU Assignment forum on Learn](#).

Downloading the notebook

Download the Jupyter notebook, `assignment.ipynb`, and the data files, `imudata.csv` and `robot.csv` from the ENMT301 Learn page.

Running the notebook on your computer

There are two options for running the notebook on your own computer:

1. Install anaconda (<https://www.anaconda.com>) and view the notebook running jupyter. Anaconda is an open-source packaging of common python packages for data science.
2. Install Microsoft vscode (<https://code.visualstudio.com>) with the python extension and load the notebook. Microsoft vscode is a free open-source IDE. You will also need to install the Python packages NumPy, SciPy, and Matplotlib using pip.

Running the notebook on a lab computer

Anaconda and Microsoft vscode are installed on the computers in the Mechatronics and CAE labs.

Remote access to lab computers

You can also access the CAE computers remotely:

1. Connect with web browser to <https://gostudent.canterbury.ac.nz>
2. Log in with uclive email address.
3. You should see the ITS workroom and CAE workroom.
4. Click on CAE workroom icon to log in to a CAE computer.

Online Jupyter notebooks

Notebooks can be run online at <https://jupyter.org/try-jupyter/lab/>

Instructions

Follow the instructions in the report template report-template.pdf. You will need to run all cells (Shift+Enter) in the Jupyter notebook. You should then see a number of plots.

Report

Create a PDF document using either the Word template report-template.docx or the LaTeX template report-template.tex.

Zip your PDF document and code into a single file and submit to Learn.

You will be marked on the quality of your observations, plots, and writing. You will not be marked on your code.

The report is due 5 pm July 24. As set out in the ENMT301 Course Outline: standard departmental (Mechanical) late penalties shall apply:

- -20% of your achievable assignment grade 0-24 hours late
- -20% of your achievable assignment grade for each 24 hours thereafter

AI tools

AI tools are permitted for certain parts of this assessment:

1. Improving your writing and providing feedback for editing.
2. Gathering and summarising knowledge on a subject that is outside your expertise.
3. Improving your software code and providing feedback on errors and implementations.